Enhancing Few Shot Out of Scope Detection with Contrastive Learning

Extension of the CLINC150 OOS Detection Benchmark

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Citation

“A new approach for fine-tuning sentence transformers for intent classification and out-of-scope detection tasks”

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Background

Task: Intent classification with out of scope (OOS) detection

CLINC150 Dataset: 150 intents across diverse domains

Challenge: Real world systems encounter OOS intents

Goal of Original Paper: Learn in scope representations and train for OOS detection.

Motivation

Motivating Problem: Real world models often face data scarcity in low resource settings.

Our Focus: Can we preserve OOS detection performance in few shot settings?

Research Question: Can contrastive learning improve generalization and separation under few shot conditions?

Original Paper’s Implementation

Model: BERT + Max Pooling + linear classifier

Loss: Cross Entropy

OOS Accuracy Metric: Embedding distance from class means

Results:

Test Accuracy: 96.1%

Test AUPR (OOS): 0.9191

Test AUROC (OOS): 0.9779

Original Paper's Implementation in Few Shot Scenario

Setup: 5 examples per class (750 total training examples) with same model and training procedure

Results:

Test Accuracy: 79.8%

Test AUPR (OOS): 0.5293

Test AUROC (OOS): 0.8364

Observation: Drop in performance as expected due to limited supervision

Few Shot Scenario + Contrastive Learning Implementation

Added Component: Supervised Contrastive Loss

Results:

- Test Accuracy: 78%
- Test AUPR (OOS): 0.6017
- Test AUROC (OOS): 0.8761

Improvement: Slight OOS detection gain over few shot baseline

Tradeoff: Small drop in in-scope classification

Summary of Results

Set up	Test Acc	Test AUPR	Test AUROC
Original Paper	96%	0.92	0.98
Original Paper + Few Shot	80%	0.53	0.84
Few Shot + Contrastive loss	78%	0.60	0.88

Key Take Aways

Few shot learning reduces both in scope and OOS performance

Contrastive learning helps OOS separation, even in low data settings

However, classification reduces slightly as a tradeoff.

Next Steps

Tune contrastive loss weights

Try alternative pooling methods

Experiment with Prototypical Networks